

Privacy-Preserving Anomaly Detection With Homomorphic Encryption for Industrial Control Systems in Critical Infrastructure

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Abstract—Critical infrastructure (CI) is essential for societal and economic stability, making it a prime target for cyber threats. Traditional anomaly detection models like LSTM and Transformers require substantial computational resources, which are often unavailable in CI environments. Cloud computing offers on-demand resources but introduces privacy concerns due to the need to transmit sensitive data to cloud servers. Homomorphic encryption (HE) enables secure processing of encrypted data but is computationally intensive, particularly due to operations like bootstrapping. This letter proposes a bootstrapping-free lightweight anomaly detection model optimized for homomorphically encrypted data, leveraging CI's operational characteristics. The model employs a two-stage data separation process and introduces state-vectors for normal operation detection, forming an allowlist anomaly detection approach. Experimental results on the SWaT and WADI datasets demonstrate the model's competitive performance and efficiency, with significantly reduced training times while maintaining robust security.

Index Terms—Anomaly detection, critical infrastructure (CI), homomorphic encryption (HE), industrial control system (ICS), privacy-preserving machine learning.

I. INTRODUCTION

CRITICAL infrastructure (CI), encompassing essential systems, such as power grids and water treatment facilities, serves as a foundational element of societal and economic stability [1]. With the ongoing expansion of Internet connectivity, these infrastructures face increasing vulnerability to sophisticated cyber threats, exemplified by incidents like Stuxnet [2] and the Colonial Pipeline ransomware [3]. The paramount importance of CI necessitates a strong focus on anomaly detection over preventative measures, as availability in these systems is crucial [4]. Consequently, substantial research efforts have been directed toward developing robust anomaly detection techniques that utilize time-series data [5], [6], [7]. However, the resource-intensive nature of these methods poses a challenge for deployment within CI environments, where computing resources are often limited.

To address these constraints, cloud computing has been proposed as a viable solution, offering scalable, on-demand

computational resources that facilitate the efficient deployment of complex models. However, transferring data to third-party cloud providers poses significant security risks, particularly for CI that manages highly sensitive information. Since any data leakage within CI can disrupt operations, compromise security, and intensify hybrid threats, which pose significant risks to its stability and resilience, it is essential to ensure the protection of CI data [8]. While traditional encryption methods can secure data, they typically do not allow for arbitrary computations on encrypted data. To address this limitation, advanced privacy-preserving techniques, such as differential privacy [9], secure multiparty computation [10], and homomorphic encryption (HE) [11] have been developed. Among these, HE is particularly promising for machine-learning-based anomaly detection, as it enables direct computations on encrypted data, thus safeguarding sensitive information throughout the analysis process.

Fully homomorphic encryption (FHE) supports an unlimited number of additions and multiplications on encrypted data [12]. Recent advancements, such as the CKKS scheme [13], have introduced the capability to perform floating-point operations, making FHE suitable for applications in deep learning models. However, FHE remains computationally intensive, particularly when applied to large datasets or ML models requiring numerous multiplications. For example, applying FHE to a convolutional neural network for image processing can take approximately three hours to process a single image [14]. This is primarily due to the bootstrapping process, which becomes necessary after multiple multiplications and significantly drives up both computational costs and time.

Therefore, there is a need for lightweight models that can efficiently operate on homomorphically encrypted data without the significant overhead introduced by bootstrapping. In this letter, we propose a two-stage data separation method that leverages the unique characteristics of industrial control systems (ICS) in CI to learn their normal behaviors using lightweight, bootstrapping-free models. This approach facilitates efficient intrusion detection with reasonable inference times, while maintaining both high security and performance. Our contributions are as follows.

- 1) To the best of our knowledge, this is the first work applying HE to ICS in CI for anomaly detections.
- 2) We propose a privacy-preserving lightweight model that utilizes the characteristics of ICS in CI, allowing intrusion detection system (IDS) to be performed without bootstrapping. Our model demonstrates comparable performance with recent works.
- 3) Our model exhibits reasonable inference time with the HE, underscoring the applicability of HE and the methods for its implementation.

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By addressing the challenges associated with HE in CI, our research offers a practical and efficient solution for enhancing industrial security while maintaining data privacy.

II. RELATED WORK

A. Homomorphic Encryption Schemes

Homomorphic encryption (HE) is a public-key cryptography method that allows computations on encrypted data without decryption, with results matching those from plaintext operations once decrypted. HE schemes are categorized by their supported operations and limitations [15]: Partially HE supports either addition or multiplication, Somewhat HE (SHE) supports both but with limited computations (circuit depth), Leveled HE extends SHE by adjusting parameters to control circuit depth, and FHE supports both operations with arbitrary circuit depth, mitigating error accumulation through bootstrapping, though it is computationally expensive.

Recent FHE advancements include the Cheon–Kim–Kim–Song (CKKS) scheme [13] and FHE over the torus (TFHE) [15]. While TFHE handles Boolean operations efficiently, CKKS supports approximate arithmetic on real numbers and vector operations, making it suitable for ML. However, CKKS requires rescaling after each multiplication, and repeated multiplications eventually necessitate computationally intensive bootstrapping. Though HE has been applied to privacy in ML [14], IDS in constrained environments is still underexplored. We address this by using CKKS to develop a bootstrapping-free IDS through optimized multiplication management.

B. Anomaly Detection on Time Series Data

Recent advancements in time series anomaly detection have shown high detection rates using graph- and transformer-based models. For instance, in graph deviation networks (GDN) [6], replacing all activation functions with square activations results in 15 multiplications. While this avoids bootstrapping, the inference time remains impractically long due to the embedding vector size and graph edges. Similarly, the anomaly transformer [7] requires substantial multiplications during the self-attention process, leading to prolonged inference times as data dimensionality increases. In addition, the use of exponential functions, which are incompatible with HE, further complicates implementation. Therefore, practical applications must consider feasible multiplications in HE, the precision of approximate activation functions, and data volume.

Even lightweight anomaly detection models are not optimized for encrypted environments, leading to significant computational overhead. For instance, an LSTM-based autoencoder [16] requires polynomial approximations for activation functions to support HE. Using a cubic approximation [17] for the sigmoid function, a lightweight model demands 48 multiplications, which necessitates bootstrapping and still suffers from prolonged inference times due to large input data size. This indicates the impracticality of using HE in such scenarios.

III. PRIVACY-PRESERVING ANOMALY DETECTION MODEL

A. Overview

This research proposes a lightweight model for anomaly detection in ICS using HE. The model employs a two-stage data separations as illustrated in Fig. 1: first, dividing

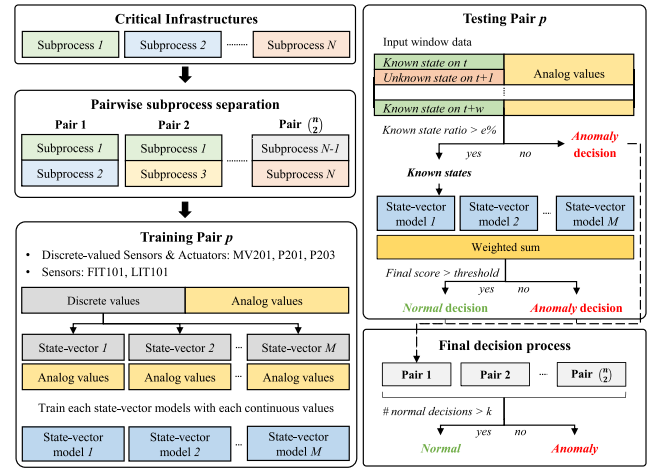


Fig. 1. Proposed anomaly detection system process.

the whole process into pairwise subprocesses (referred to as *pair*), and then refining these into *state-vectors* defined by discrete-valued sensors and actuators, which exhibit regular and cyclic patterns. An autoencoder-based network is designed for each state-vector to learn the normal state representation, functioning as an allowlist for anomaly detection. The normal decisions of each pair are aggregated to define the overall system state. The training protocol leverages cloud computing resources, ensuring data privacy through homomorphically encrypted data exchange between the client and server. This approach balances model efficiency and security for reliable intrusion detection in CI.

B. Model Design

To achieve model lightweighting for HE, data dimensions and representations should be reduced. This is accomplished through two-stage data separation and the implementation of an autoencoder-based ICS anomaly detection system.

1) *Two-Stage Data Separation*: CIs are typically composed of multiple interconnected subprocesses. Processing all data simultaneously increases both data volume and computational costs. Furthermore, attackers often exploit this by targeting either individual or multiple subprocesses simultaneously. If a single subprocess is compromised, anomalies may propagate to connected downstream subprocesses. To address this, we propose separating the datasets into pairwise subprocess units, preserving the inherent characteristics of the CI while minimizing computational overhead. For an infrastructure with N subprocesses, this approach generates $\binom{N}{2}$ unique pairs, enabling a more focused analysis of interdependencies.

Directly learning from each data pair may result in sub-optimal representations when using lightweight models. To enhance efficiency, we introduce state-vectors to segment the data, thereby reducing complexity. Under normal operating conditions, CI data typically exhibits low entropy, leading to regular and cyclic patterns in both discrete and analog values in ICS. A feedback loop between the discrete values controlling the system and the analog values reflecting its state makes it ideal to define state-vectors using discrete-valued sensors and actuators. By leveraging these state-vectors as a basis for system state estimation, the data are separated into segments that consist of analog values corresponding to stable state

conditions. This two-stage segmentation effectively captures complex system dynamics with lower computational costs.

For each dataset segmented by state-vectors, an allowlist model learns normal data patterns within windowed segments. The proposed model consists of N blocks, each comprising a fully connected layer, a square activation function, and a layer normalization layer. Due to the inefficiency of typical activation functions under HE, polynomial approximations are used instead, resulting in $3N$ multiplications. This design enables HE implementation without requiring bootstrapping.

The trained allowlist models assess the pair's status by evaluating the proportion of unknown state-vectors within a window. If the proportion exceeds a predefined threshold, the pair is classified as unknown. Otherwise, the window's status is estimated using the weighted sum of reconstruction losses from each state-vector model. Given CI's regular and cyclic operations, this anomaly detection method, based on the proportion of unknown state-vectors, is highly effective.

2) *Allowlist Anomaly Detection*: The model generated from the state-vectors of pairs learns only the normal states, allowing it to function as an allowlist. If a state-vector model detects normal state, the corresponding pair is deemed normal. For $\binom{N}{2}$ pairs, if k pairs detect normalcy, the system is considered normal. Conversely, if the majority of pairs do not detect normalcy, the system is considered attacked.

C. Training Protocol

In ML, backpropagation is crucial for updating parameters. Encrypted weights increase depth consumption with data size and epochs, requiring frequent bootstrapping. This letter proposes a protocol to avoid bootstrapping, with the CI owner as the client and the cloud service provider as the server. The protocol proceeds as follows.

- 1) The client transmits encrypted data to the server.
- 2) The server trains the model on this encrypted data, maintaining encryption throughout.
- 3) The server requests the client to compute functions, such as gradients and loss values.
- 4) The client decrypts the data, evaluates the functions, and returns the values to the server unencrypted.
- 5) The server uses these values to perform backpropagation and update the model's parameters.

IV. EXPERIMENTS

A. Metrics

We evaluate the model's performance using precision, recall, and F1 score. Precision measures the accuracy of detected anomalies, recall assesses the system's ability to identify all actual anomalies, and F1 score provides a balanced evaluation by combining precision and recall.

- 1) Precision = $TP/(TP + FP)$.
- 2) Recall = $TP/(TP + FN)$.
- 3) F1 score = $2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall})$.

B. Dataset

There are few publicly available datasets that represent real-world scenarios. In this study, we evaluate our model using two widely recognized datasets: 1) secure water treatment (SWaT) and 2) water distribution (WADI). The SWaT dataset consists of time-series data from 51 sensors and actuators recorded every second, organized into six subprocesses [18].

TABLE I
PERFORMANCE COMPARISON ON SWaT AND WADI DATASETS

Method ([6], [7], [16])	SWaT			WADI		
	Prec	Rec	F1	Prec	Rec	F1
PCA	24.92	21.63	0.23	39.53	5.63	0.10
KNN	7.83	7.83	0.08	7.76	7.75	0.08
FB	10.17	10.17	0.10	8.60	8.60	0.09
AE	72.63	52.63	0.61	34.35	34.35	0.34
DAGMM	27.46	69.52	0.39	54.44	26.99	0.36
LSTM-VAE	96.24	59.91	0.74	87.79	14.45	0.25
MAD-GAN	98.97	63.74	0.77	41.44	33.92	0.37
GDN	99.35	68.12	0.81	97.50	40.19	0.57
AnomalyTrans	91.55	96.73	0.94	-	-	-
LW-LSTM-VAE-M	89.88	79.65	0.84	72.71	31.25	0.43
Ours	99.09	68.26	0.81	62.83	54.59	0.58

WADI, an extension of SWaT, includes 127 elements and is structured into eight subprocesses [19]. Both datasets contain attack data with ground-truth labels. The generated state-vectors successfully cover 99% of each dataset, ensuring comprehensive evaluation.

C. Preprocessing

Given the heterogeneity of the dataset, which may hinder model training, min-max normalization is applied based on the training data. The normalized data are then segmented into state-vectors and processed using a windowing technique to capture temporal context effectively. The optimal window size was determined experimentally to be 30 for SWaT and 60 for WADI. If a state-vector's duration is shorter than the window size, the dataset is excluded, as such instances have minimal impact. For evaluation, any window containing an attack is labeled as an attack accordingly.

D. Training Environments

To optimize HE utilization without bootstrapping, the proposed model is configured with four layers, with hidden dimensions set to $30 \times$, $3 \times$, and $30 \times$ the number of input features, respectively. For CKKS encryption, a minimum security level of 128 bits is ensured, adhering to current standards for secure applications. Considering the 12 required multiplications during evaluation, the polynomial modulus degree is set to 16384 , with a coefficient modulus of 428 bits ($[40] + [29] \times 12 + [40]$) and a scale factor of 40 bits. The detection threshold for identifying unknown states is defined as 50%, while the voting threshold k for system state determination is set to 10 for the SWaT dataset and 19 for the WADI dataset.

The model is initially trained in plaintext and then evaluated with HE to ensure compliance with a predefined similarity criterion. Results show that the cosine similarity between the plaintext and HE results reaches 99%, indicating high consistency. The implementation is developed in Python, utilizing the TenSEAL library for HE operations. All experiments were conducted on a machine equipped with an AMD Ryzen 9 3900X processor and an NVIDIA GeForce RTX 3090 GPU.

E. Results

Despite the inherent constraints of lightweight models, our approach achieves performance comparable to recent studies, as shown in Table I. The two-stage separation using discrete value-based state vectors shows strong performance on both

TABLE II
INFERENCE TIMES AND PERFORMANCE OF SOTA MODELS AND OURS

Method	SWaT			WADI		
	Window	Inference	F1	Window	Inference	F1
MAD-GAN	(150,5)	1,005s	0.77	(240,5)	1,116s	0.37
GDN	(5,51,64)	21,868s	0.81	(5,127,128)	75,590s	0.57
Ours	(30,15)	601s	0.81	(60,37)	2,072s	0.58

the SWaT and WADI datasets, even with HE applied for privacy preservation. For SWaT, high precision and adequate recall indicate that the state-vector model effectively functions as an allowlist. In contrast, for WADI, higher recall and lower precision suggests that robust rejection of unknown states improves anomaly detection, highlighting a key tradeoff: while HE preserves data privacy, the multiplication limits required to avoid bootstrapping impact efficiency. As noted in Related Work, state-of-the-art (SOTA) methods, including AnomalyTrans and LSTM-based models, though effective, are impractical for HE integration. Our model, however, achieves results comparable to SOTA methods like MAD-GAN and GDN through data reduction leveraging CI characteristics.

In addition to the number of multiplications, data volume plays a crucial role in determining the computational requirements of HE. Table II compares the inference times of previous works, calculated using the window sizes reported in their studies, excluding any HE-related overhead. For SWaT, graph-based models like GDN require fewer multiplications but suffer from increased computational complexity due to large embedding dimensions, making them inefficient under HE. Conversely, MAD-GAN reduces inference time through feature selection, yet its suboptimal performance on WADI indicates over-reliance on specific features. Our approach, utilizing a moderate window size and CI subprocess-based data separation, achieves robust performance across both datasets, demonstrating adaptability in diverse ICS environments.

V. DISCUSSION AND CONCLUSION

The final inference time of 601 s in SWaT exceeds the window size of 30, making it unsuitable for online anomaly detection, which demands inference times under 1 s. Without HE, our model achieves an inference time of less than 1 millisecond, highlighting the substantial computational cost of HE. In CI, where availability often outweighs immediate attack responses, offline detection remains viable. Our method proves feasible even with HE, outperforming existing studies where excessive inference times hinder offline detection.

The proposed lightweight anomaly detection model demonstrates the practical applicability of HE and comparable performance on both the SWaT and WADI datasets. By utilizing HE without bootstrapping, the model effectively balances performance and privacy. Through a two-stage data separation process, it significantly reduces data volume and multiplications, achieving inference times approximately

thirty-six times faster than GDN. Although the reduction of multiplications and avoidance of bootstrapping introduce certain tradeoffs, our model still achieves results on par with SOTA methods, highlighting its suitability for CI deployment. Future work will focus on optimizing performance, refining the state vector, and maintaining robust security to enable online detection with HE. In addition, we will investigate the potential to further enhance performance by incorporating minimal bootstrapping.

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